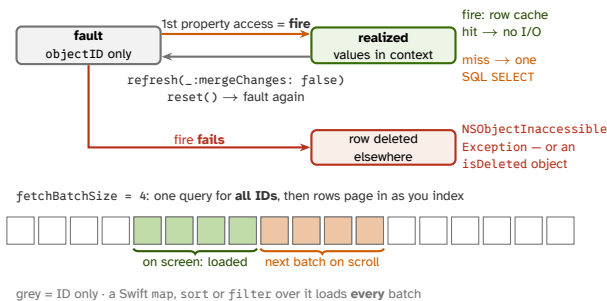


In one line: Fetched objects are **faults**, filled on first touch from the row cache or a SQL SELECT — performance = few round-trips. **Batch requests** run as SQL on the store: fast, but no context sees them until you **merge the returned IDs**. **Persistent history** logs every transaction (who, what) in the store, so each context, process and extension catches up from its own **token**.

A fault's life — and the batch-size window



Fetch knobs

Knob (default)	Effect · use
<code>returnsObjectsAsFaults (true)</code>	false: filled now; relationships still faults
<code>includesPropertyValues (true)</code>	false: IDs only, no row cache — deletes IDs first, rows in pages — long lists, FRC
<code>fetchBatchSize (0 = all)</code>	one query per key path, not a fault per object: kills N+1
<code>relationshipKeyPathsForPrefetching</code>	read-only dicts, no objects; + <code>NSExpressionDescription = SUM/AVG in SQL</code>
<code>propertiesToFetch + .dictionaryResultType</code>	<code>SELECT COUNT · LIMIT</code>
<code>count(for:).fetchLimit</code>	

A fetch **always runs SQL**; its faults then fire from the row cache it filled. Big import: `save + reset()` every N rows.

Batch requests — SQL on the store (SQLite only)

Request	iOS	Notes · result type for IDs
<code>NSBatchUpdateRequest</code>	8	<code>propertiesToUpdate = one UPDATE</code> <code>.updatedObjectIDsResultType</code>
<code>NSBatchDeleteRequest</code>	9	basic delete rules; Deny unsupported <code>.resultTypeObjectIDs</code>
<code>NSBatchInsertRequest</code>	13	dicts or a handler; unique constraints checked; no relationships (<i>unverified</i>) <code>.objectIDs</code>

All skip the context: no validation, no `willSave`, no notifications; objects in memory go stale. Merge the IDs: class method `mergeChanges(fromRemoteContextSave:into:)` — or let history do it.

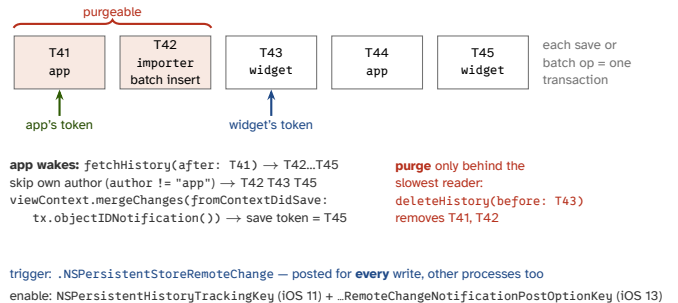
Example — FRC + diffable, and a batch delete the UI sees

```
let r = NSFetchedRequest<Item>(entityName: "Item")
r.sortDescriptors = [NSSortDescriptor(key: "date", ascending: false)]
r.fetchBatchSize = 30
frc = NSFetchedResultsController(fetchRequest: r, managedObjectContext:
    viewContext, sectionNameKeyPath: nil, cacheName: nil)
frc.delegate = self; try frc.performFetch()
func controller(_ c: NSFetchedResultsController<NSFetchRequestResult>,
    didChangeContentWith s: NSDiffableDataSourceSnapshotReference) {
    dataSource.apply(s as NSDiffableDataSourceSnapshot<String, NSManagedObjectID>)
}
// inside bg.perform: 50k rows gone in one DELETE, then tell viewContext
let del = NSBatchDeleteRequest(fetchRequest: NSFetchedRequest(entityName: "Item"))
del.resultType = .resultTypeObjectIDs
let res = try bg.execute(del) as? NSBatchDeleteResult
let ids = res?.result as? [NSManagedObjectID] ?? []
NSManagedObjectContext.mergeChanges(fromRemoteContextSave:
    [NSDeletedObjectsKey: ids], into: [viewContext]) // FRC animates it
```

NSFetchedResultsController

- Watches **one context** for one fetch request (≥ 1 sort descriptor; `sectionNameKeyPath` follows the first sort key). It moves only when a change lands *in that context* — a save there or a merge.
- iOS 13 controller(`_didChangeContentWith:`): a snapshot of `NSManagedObjectContext` (sections = `String`);

Persistent history — the store's changelog



implement it and the per-row callbacks stop. A `CollectionDifference` variant exists too.

- An in-place edit keeps the same ID, so the diff sees *nothing*: `reconfigureItems` (iOS 15) the updated IDs, or let cells observe their object.

Persistent history

- Each `NSPersistentHistoryTransaction`: token, timestamp, author, contextName, changes (object ID, insert/update/delete, updatedProperties). Name writers: `ctx.transactionAuthor = "importer"`.
- Read on a background context with `NSPersistentHistoryChangeRequest.fetchHistory(after:)`; persist the last token (`NSSecureCoding`) per reader.
- Purge with `deleteHistory(before:)` a token, date or transaction. Once tracked, opening the store *without* the key makes it **read-only**.

App + extension on one store

Store URL in the App Group container; both targets set both keys on the store description **before** `loadPersistentStores`. Each process has its own coordinator, so `automaticallyMergesChangesFromParent` never sees the other — each reader keeps its own token in the group and merges foreign authors.

Debug — launch arguments

`-com.apple.CoreData.SQLDebug 1` (up to 4): every SQL statement + time — N+1 shows as a SELECT per row.
`...ConcurrencyDebug 1`: crash on wrong-queue access.
`...MigrationDebug 1`: migration steps.

Traps

- ! FRC stale after an import: the change never reached its context (no merge, a batch request, another process).
- ! Batch delete, UI still shows the rows — nobody merged the IDs.
- ! Purging on a timer before the widget read it: it silently misses changes.
- ! Reading objectID or `isFault` does not fire a fault.

Remember

Faults are free until touched; batches are fast until forgotten; history is read after your token and purged behind the slowest reader.